1. Koszul complex

We give two formulations of Koszul, one for proving it to be a min. res.; one for computations & intuitions.

① For $x \in R$, $Kos(x) :=$

$$0 \to R \xrightarrow{x} R \to 0,$$

and $Kos(\underline{x}) = Kos(x_1, \cdots, x_{n-1}) \otimes Kos(x_n)$.

i) Equivalently, it's the total complex of the double complex:

$$\begin{array}{ccccccccc}
\cdots & \to & R^{\binom{n}{k}} & \cdots & \to & R^{\binom{n}{2}} & \xrightarrow{\partial_2} & R^n & \xrightarrow{\partial_1} R \to 0 \\
 & & \downarrow (-1)^k x_{n+1} & & & \downarrow x_{n+1} & & \downarrow -x_{n+1} & \downarrow x_{n+1} \\
\cdots & \to & R^{\binom{n}{k}} & \cdots & \to & R^{\binom{n}{2}} & \xrightarrow{\partial_2} & R^n & \xrightarrow{\partial_1} R \to 0
\end{array}$$

Note that $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$, so $F_k = R^{\binom{n}{k}}$, all the same.

ii) Total complex tells us that if we let $C := Kos(x_1, \cdots, x_n)$, then:

$$[Kos(x_1, \cdots, x_{n+1})]_i = C_{i-1} \oplus C_i$$

✡✡ Clearly from the double complex, we have:

$$\partial_i := \begin{pmatrix} \partial_C & 0 \\ (-1)^{i-1} x_{n+1} & \partial_C \end{pmatrix}, \qquad \begin{array}{ccc} \oplus & & C_{i-1} \xrightarrow{\partial_C} C_{i-2} \\ & & \downarrow (-1)^{i-1} x_{n+1} \\ C_i & \xrightarrow{\partial_C} & C_{i-1} \end{array} \oplus$$

Thus:

✡✡ $Kos(x_1,\cdots,x_{n+1}) = cone\left(Kos(x_1,\cdots,x_n) \xrightarrow{x_{n+1}} Kos(x_1,\cdots,x_n)\right)$.

(using the convention that C_{n-1} is before C_n).

(iii) ✡ Thm: (R,m), $f_1,\cdots,f_n \in m$. TFAE:

a) $f_1,\cdots,f_n$ reg. seq.

b) $Kos(f_1,\cdots,f_n)$ is min. res. of $R/(f_1,\cdots,f_n)$ i.e. $H_i(Kos(\underline{f})) = 0 \ \forall i>0$.

c) $H_1(Kos(\underline{f})) = 0$

Pf: a) $\Rightarrow$ b): we prove by induction.

• $i=1$: $\quad 0 \to R \xrightarrow{f_1} R \to R/(f_1) \to 0$

is min. free.

• $i=n$: We have SES $0 \to C \to K \to \Sigma C \to 0$,

where $C = Kos(f_1,\cdots,f_n)$, $K = Kos(f_1,\cdots,f_{n+1})$.

$\exists$ LES:

$$\cdots \to 0 \to H_1(K) \to H_0(C) \xrightarrow{\cdot x_{n+1}} H_0(C) \to H_0(K) \to 0 \to 0$$

Since x_{n+1} is regular, $H_1(K) = 0$.

$\Rightarrow H_n(K) = 0 \ \forall n>0$.

b) $\Rightarrow$ c): trivial.

c) $\Rightarrow$ a): We use the same LES as in "a) $\Rightarrow$ b)".

Then $C \xrightarrow{\cdot x_{n+1}} C$ is injective, so x_{n+1} is regular.

✡ Then $Kos(\underline{f})$ is minimal res. since the entries of ∂_i are $\pm f_i \in m$.

② $Kos(\underline{f})$ is the exterior algebra generated by $\underline{f}$; so it's graded comm., and we usually write:

$$Kos(\underline{f}) = R[e_1, \cdots, e_n : \partial(e_i) = f_i],$$

obtained by R adjoining e_i that kills cycles f_i in the homology.

Thus $K_m = \bigoplus_{1 \le i_1 < \cdots < i_m \le n} R e_{i_1} \cdots e_{i_m}$, and $\partial(e_{i_1} \cdots e_{i_m}) = \sum_{k=1}^{m} (-1)^{k-1} f_{i_k} e_{i_1} \cdots \hat{e}_{i_k} \cdots e_{i_m}$, extended by Leibniz rule.

✡ Properties of $Kos(\underline{x})$: $I = (\underline{x}) \triangleleft R$, $K := Kos(\underline{x})$.

i) $H_i(K) = 0$ for $i < 0$ or $i > n$; $H_0(K) = R/I$, $H_n(K) = ann(I)$.

ii) $H_i(K)$ is fg.

iii) $H_i(K)$ is killed by I.

Pf: We prove iii). For any $x_m \in I$, consider $s_i : K_i \to K_{i+1}$,

$$s_i(e_{k_1} \cdots e_{k_i}) = e_m e_{k_1} \cdots e_{k_i}$$

Then $\partial s_i(e_{k_1} \cdots e_{k_i}) = x_m e_{k_1} \cdots e_{k_i} - e_m \partial(e_{k_1} \cdots e_{k_i})$.

$$s_{i-1}\partial(e_{k_1} \cdots e_{k_i}) = e_m \partial(e_{k_1} \cdots e_{k_i}).$$

$\Rightarrow \partial s_i t s_{i-1}\partial = x_m$, and this extends by linearity

$\Rightarrow$ so is $a \in I$, $a = \sum b_m x_m$.

Thus $a \cdot -$ is nullhomotopic $\Rightarrow H_i(K)$ killed by I.

2. DGAs

① Facts about DGAs & dg ideals.

To check: $\begin{cases} \text{DGA: } \partial(ab) = \partial(a)b + (-1)^{|a|}\partial(b) \\ \text{Graded comm.: } ab = (-1)^{|a||b|}ba \\ \text{Map of dga: } f\partial = \partial f \ \& \ f(ab) = f(a)f(b) \\ \text{dg ideal: } \partial(I) \subset I. \end{cases}$

i) For $F_\cdot$ free res. with algebra structures & $M = R/I$, we have a quasiisom. of dgas: $F \simeq R/I$

ii) For $Kos(f)$, $J = (e_1 \cdots e_n, f_1, \cdots f_n)$ is a dg ideal <u>generated</u> by $e_1, \cdots e_n$.

iii) $A \twoheadrightarrow A/I$ for dg ideal I.

- A/I is dga, $A \twoheadrightarrow A/I$ dga map.

pf: Well-def. since I closed under $\cdot$ & ∂, each degree: $(A_n + I)/I$.

- $A \simeq A/I \Leftrightarrow H(I) = 0$.

$$\begin{array}{ccccc} A_{n+1} & \xrightarrow{\partial} & A_n & \xrightarrow{\partial} & A_{n-1} \\ \downarrow & & \downarrow & & \downarrow \\ (A_{n+1}+I)/I & \xrightarrow{\partial'} & (A_n+I)/I & \xrightarrow{\partial'} & (A_{n-1}+I)/I \end{array}$$

Suffices to prove injectivity.

If $z \in \ker \partial'$ & $z = \partial'(y)$, needs to prove $z = \partial(y')$ for $y' \in A_{n+1}$.

Now $z = \partial'(y'+i)$ for $i \in I$, so $z = \partial(y') + \underbrace{\partial(i)}_{\in I}$, so z can also be represented by $\partial(y')$, and thus $z' = \partial(y')$.

↓ as an elt. in H_n.

② Local dga.

i) Local dga is dga with unique max. dg ideal.

ii) e.g. • $Kos(f)$ is a local dga with $m_A = m_R \oplus A_{\geq 1}$

• More generally, if $A_0 = R$ local, then A is local with $m_A = m_R \oplus A_{\geq 1}$

iii) $\varphi : A \to B$ local map of dga if $\varphi(m_A) \subset m_B$.

③ Hom Complex

i) Def: Let F, G be complexes of R-mod, define the complex

$$\mathrm{Hom}_R(F,G)_n := \prod_i \mathrm{Hom}_R(F_i, G_{i+n})$$

collection of deg. n R-lin. maps

with differential:

$$\partial(\varphi) = \partial_G \circ \varphi - (-1)^{|\varphi|} \varphi \circ \partial_F.$$

ii)

- $Z_0(\mathrm{Hom}_R(F,G))$ are chain maps.
  $\partial(\varphi) = 0 \iff \partial_G \circ \varphi = \varphi \circ \partial_F \iff \varphi$ chain map
- $B_0(\mathrm{Hom}_R(F,G))$ are nullhomotopic maps.
  $\exists s$ st. $\partial_G s + s\partial_F = \varphi \Rightarrow \varphi$ nullhomotopic.
- Thus $H_0(\mathrm{Hom}(F,G))$ are maps up to homotopy.

iii) $\mathrm{End}_R(M) = \mathrm{Hom}_R(M,M)$ is dga.

④ dg module.

i) A dg module M over an $(R-)$ dga A is a graded module s.t.

$$\star \quad \partial(am) = \partial_A(a)m + (-1)^{|a|} a\,\partial(m)$$

ii) $(\Sigma C)_n = C_{n-1}$, $\partial_{\Sigma C} = -\partial_C$, so is $\Sigma^k C$ and $\Sigma^{-1} C$.

iii) One can verify $\Sigma^l C$ is dg module with:

$$a \cdot \Sigma^l m := (-1)^{|a|l} \Sigma^l (am).$$